PROJECT PLAN / 01

Urbanova

See what new housing could change in your neighbourhood

Neighbourhood scale, seasonal sunlight, streetcar readiness and AI research.

Urbanova helps Toronto residents picture and understand how proposed residential developments along the westbound TTC 505 Dundas streetcar corridor could affect their community: the buildings they see, the sunlight in places they use, and the transit they depend on. Transit simulation remains the core of the product, empowering residents and planners to explore how a commute could change and test service improvements.

Our central question and core decision

How could this development change my commute and neighbourhood, and what improvements would help accommodate that growth? For the same planned homes, what transit service needs to be ready at move-in, what resources does it require, and how sensitive is the answer to demand and delivery timing?

Three connected ways to explore change, plus AI research

1. Neighbourhood & transit workspace. Explore a navigable 3D neighbourhood from eye level (A. Street-Level View), compare existing and added/removed ground shadows by date and time (B. Sunlight & Shadows), and follow simulated streetcar crowding, queues, waits and boarding chances along 505 Dundas westbound with service package comparisons (C. Aerial View & Transit).

2. Papiers research partner. Support questions such as 'Why does my stop get worse even though the new building is farther up the route?' Read research context, notice patterns in computed results, propose a hypothesis, run a controlled simulation experiment, evaluate it, and select a next investigation redirected by human feedback.

The resident experience (user story)

Existing residents are the primary audience, alongside community groups, prospective residents, journalists and planners. A resident opens the 505 Dundas neighbourhood, selects a real proposal (239, 500 or 761 Dundas St E), toggles the translucent 3D building masses from street level and above, and compares seasonal shadows at a familiar park or street corner. Next, they inspect their usual westbound stop across two occupancy phases, test whether extra service or transit priority relieves crowding within capital, operating and fleet limits, and use the Papiers partner to investigate a downstream bottleneck before saving an evidence-linked brief.

Benefit and intended distinction

Urbanova's intended distinction is a resident-centred transit simulation connected to real Toronto housing proposals, familiar places and seasonal sunlight changes, with adjustable scenarios and an inspectable AI research trail that help people understand their commute, test improvements, and bring understandable evidence into community discussions.

Hack the Valley 11 | Planning version 2: October 17, 2026. Updated for the Urbanova resident-centred focus in project_desc.md. Flags F1-F7 carry review concerns: amber marks open items; blue marks plan decisions.

PROJECT PLAN / 02

Scope and product features

Three connected neighbourhood views and one complete transit-and-research workflow

Area	Committed target
Study area	TTC 505 Dundas streetcar, westbound, Broadview Ave to TMU Station at Yonge/Dundas; 9 stops, 8 links and three selected proposals (239, 500 and 761 Dundas St E).
3D & sunlight	Navigable street-level and aerial 3D views with translucent proposal toggles (2,444 existing buildings, 4 proposal tower masses) and date/time shadow comparison (today, added, removed).
Time and demand	Two occupancy phases; 07:00-09:00 AM peak arrival window; low, central and high demand; one delayed-upgrade case; expanded mapped-building residential baseline.
Packages & research	Existing service, a reviewed priority-lane option, an additional-service timetable, and both (housing stays fixed); one completed, evaluated Papiers/Gemini investigation with human redirection.
Outputs	Street/aerial 3D & shadow views, 2D map/list alternative, comparison cards, streetcar/queue replay, research history, and a saved comparison brief with assumptions and evidence.

Essential interactions

Explore. Select a proposal or stop; toggle translucent proposed towers from street level and above; inspect planning status, assumed occupancy, walking access and date/time shadow changes (both lost sunlight and summer shade). Distinguish proposed designs from simplified masses.

Compare. Edit occupancy, demand assumptions, headways, budgets and intervention dates. Show today's baseline, growth with unchanged service, and growth with each feasible package. Mark old results stale after edits.

Investigate. Ask why a stop worsens when upstream housing opens, review the agent's hypothesis and experiment, inspect evidence, add human judgment, and see the next research choice change.

Understand and save. Provide source/assumption labels, keyboard controls, readable text summaries, a 2D map/list alternative and pause for motion. Save the scenario, results and research trail together for community discussion.

OPEN F2 | Protect the complete workflow

Transit simulation remains the core of the product. Cut extra visual polish, free-roam Three.js textures and redundant AI chat before cutting the simulator, proposal/shadow comparison or Papiers research loop. If both phases cannot be completed, explicitly disclose a one-phase fallback rather than presenting it as the full target.

Outside this build

Signal priority is excluded from the core build. Flat-ground sunlight and shadows are modelled, whereas temperature and thermal comfort (heat modelling) require additional modelling and are excluded. Also excluded: arbitrary building placement, city-wide coverage, mode-choice or emissions prediction, new vehicle types, automatic code modification, and multi-agent orchestration.

PROJECT PLAN / 03

Data and demand

Real 505 Dundas geography; explicit assumptions where observations are missing

Input	Use and readiness check
Housing [3]	Three selected proposals on 505 Dundas (239, 500, 761 Dundas St E: 624, 413 and 1,197 net added units) plus expanded mapped-building residential inventory (589 buildings in simulation).
TTC GTFS [4]	Westbound 505 Dundas from Broadview (1756) to TMU Station (2136), 9 stops and 8 links on Friday 2026-10-16 (07:00-09:00 AM window, 17 scheduled trips); local bundle frozen.
Baseline riders [5]	32,100 weekday 505 boardings bound full-route context only; stop-level peak demand and incoming Broadview loads (12-24 onboard riders) are explicit scenario assumptions.
Walking, 3D & shadows [6,7]	OSM pedestrian routes (<=500 m to up to two stops), 2,444 extruded buildings + 4 proposal tower masses (buildings_3d.geojson), NOAA sun positions and ground shadows (shadows.geojson).
Packages [8]	Specify affected 505 links, travel-time effects, added streetcar dispatches, full-cycle fleet needs, capital/operating costs, availability phases and effect ranges.

Demand construction

One visible calculation chain
Net added units x occupied share x residents per occupied unit x trips per resident in the peak window x transit/corridor reachability. Then allocate those trips to accessible westbound 505 stops, times, direction and destinations.

Every factor needs a unit, denominator, source or assumption label and allowed range. Keep existing riders separate from added residents. Generate consistent passenger origins and destinations. Incoming streetcars at Broadview may already carry passengers; riders travelling beyond TMU Station remain accounted for at the segment boundary.

F1 | Corridor selected; field review and stop-level observations remain open
The corridor is fixed to TTC 505 Dundas westbound (Broadview to TMU Station, 9 stops) with 239, 500 and 761 Dundas St E supplied via simulation-data.md. Stop-level demand observations, the 761 replacement unit count, and walking entrance/crossing field review remain open; describe the build as an exploratory scenario tool, not a calibrated TTC forecast.

OPEN F5 | Fill the numerical intervention catalogue
Data and simulation owners must agree on one priority-lane scheme and one extra-service streetcar timetable, with effect ranges, capital cost, annual operating cost, price year, fleet increment and availability. Unknown values remain labelled assumptions.

Preserve evidence

Save the exact prepared input bundle, source links, retrieval dates, licences and content hashes. Version occupancy dates as scenario assumptions unless evidenced. Low/high cases vary named uncertain factors; they are not assigned real-world probabilities. No live feed or city-wide download is required during the demo.

PROJECT PLAN / 04

Simulation and decision rules

Consistent streetcar passengers, fair comparisons and visible tradeoffs

Use a discrete-event model along westbound 505 Dundas: passenger arrival, streetcar arrival, alighting, capacity-limited boarding, dwell and departure. Queues persist when streetcars are full, showing how upstream boarding leaves riders waiting farther along the route. Boarding-related dwell feeds into later vehicle spacing. Transit priority changes link travel-time/delay assumptions; extra service changes the dispatch timetable and resource requirements.

Outcome	Report
Passenger service	Mean and 95th-percentile wait; unique share denied boarding at least once; in-vehicle passenger load, remaining stop queue and unresolved passengers, by stop and rider cohort.
Journey and reliability	Within-segment ride time and wait-plus-ride time for matched origin/destination cohorts; headway variation. Incoming onboard riders at Broadview have separate segment ride-time results.
Resources	Capital cost, annual operating cost, full-cycle streetcar fleet requirement and phase availability, shown beside commute and neighbourhood outcomes.

F3 | Addressed in this plan: measure the priority option's travel benefit
Add in-vehicle and within-segment journey time alongside stop waits and vehicle crowding. Transit priority can help passengers even if scheduled headway stays constant. These measures exclude walking, transfers and travel beyond TMU Station.

Readiness and package selection

The user sets a P95-wait limit and a denied-boarding limit. A package passes only if every evaluated stop meets both limits in every selected demand case for that phase, and capital, operating, fleet and availability constraints all hold. Show aggregate passenger metrics across repeated runs plus variation between runs; a percentile is not a confidence interval.

Return the passing packages and their cost/service tradeoffs so residents and planners can explore a concrete service request. Remove dominated options from the preferred shortlist but keep them inspectable. Apply an explicit user-selected priority without adding one-time capital and recurring operating costs into an unexplained total.

OPEN F4 | Set the actual acceptance values
The rule above is the proposed default. The team must select demo thresholds and a cost priority before the first comparison. Label them user-selected, not official TTC standards. Sparse or unresolved wait observations cannot establish a pass; show an indeterminate result or failure with the outstanding queue. If nothing passes, report that outcome without reducing the homes.

Fairness and boundary checks

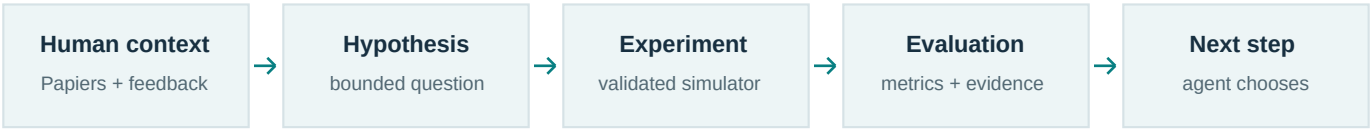
Match housing, passenger journeys and pre-generated random draws across packages. Warm up the model, measure a defined arrival cohort, then allow later service to observe its waits. At an analysis cap, retain unserved riders and wait lower bounds. Faster links never create free trips; fleet and service costs use a full round trip including return and layover.

PROJECT PLAN / 05

Papiers: the research partner

Help residents and planners investigate why a downstream stop worsens

Make this a central product capability and a target sponsor submission. It supports questions such as 'Why does my stop get worse even though the new building is farther up the route?' by extending the comparison into a research process: the agent uses source material and computed results to decide what to test, executes a bounded comparison, evaluates evidence, and chooses a follow-up question. [1,2]



Repeat within a fixed experiment budget; request human judgment when evidence is missing.

Stage	Concrete behavior and saved evidence
Read context	Retrieve a team-selected Papiers paper, question or annotation through supported access. Save source IDs, relevant excerpts and which resident or planner question informed the investigation.
Notice and hypothesize	Inspect computed stop queues, vehicle crowding, denials, headways and phase comparisons. Propose a bounded explanation, expected observable effect, controlled change and evaluation metric.
Run an experiment	Validate parameters and resource limits; freeze housing and other controls; execute paired runs through the same simulation worker. Save configuration, seeds and run IDs.
Evaluate and continue	Use computed differences and run variation to label the hypothesis supported, contradicted or inconclusive within the model. Choose a new question from the evidence, or request human expertise.
Teach back	Present the finding, failed tests, caveats and next question. Accept a human correction and record how it changes the next experiment. Write a finding to Papiers if supported.

A small experiment vocabulary

Policy tests use only the existing priority-lane, extra-service and combined packages, plus their availability dates. Diagnostic tests vary uncertain incoming Broadview loads, departure-time concentration, boarding/alighting dwell parameters or link-delay effects within declared ranges. Diagnostic changes reveal model sensitivity; they are not free or automatically actionable interventions.

F6 | Coherence decision
Papiers supplies the human research context for the experiment loop. The agent tests existing service packages and declared uncertainty ranges. A single agent with a few validated tools is enough. Its completed experiment, evaluation and next step demonstrate the research contribution.

The agent can surface overlooked consequences represented by the model, such as a downstream queue at Sherbourne or Jarvis hidden by the corridor average. For an unmodelled cause, it opens an evidence request instead of claiming to discover a real-world fact.

PROJECT PLAN / 06

A research demonstration

An illustrative investigation along westbound 505 Dundas, not a prewritten finding

Starting observation

A resident or planner compares the same housing phase across the four packages on westbound 505 Dundas. A downstream stop (such as Sherbourne or Jarvis) has frequent denied boarding even though new housing opens farther upstream near River and Parliament. A Papiers note questions the assumed load of streetcars entering at Broadview and upstream boarding concentration.

Step	What the live system must do
Hypothesis	The agent proposes that incoming occupancy at Broadview combined with upstream boarding at River and Parliament drives the downstream bottleneck. It states a measurable prediction and cites the motivating note.
Controlled test	Run lower and higher incoming-load cases with the same housing, dispatch plan and other assumptions; use matched seeds. Treat this as an uncertainty investigation, not a policy improvement.
Evaluation	Compare stop-level denial share, P95 waits, in-vehicle crowding, queues and journey time. Show actual values and run variation. If the evidence is weak or contrary, say so.
Human redirection	The user contributes a constraint: no extra streetcars are available in phase one. The agent records this and stops proposing infeasible phase-one frequency increases.
Agent-generated next step	The agent chooses a relevant remaining test from the results and updated constraint: for example, priority-effect sensitivity or upgrade delay. Its choice must respond to evidence, not replay this example verbatim.

Papiers integration contract

Minimum: a real supported read retrieves human context that changes a hypothesis or experiment. Preferred: also write a concise finding back with source IDs, run links, assumptions, limitations and next question. One direction satisfies the supplied integration requirement; both directions strengthen the feedback story. [2]

OPEN F6 | Access is the first research gate
Research owner must test the sponsor-provided MCP/API or another sponsor-approved integration path immediately. The supplied guide is marked Coming soon; authentication, available fields and read/write support are untested. Demonstrate one real operation before building against assumed endpoints. A mock or manually copied note may support development, but is not presented as live Papiers integration.

A bounded autonomous loop

A user starts an investigation. The agent may run up to three experiment batches within a wall-time and API-cost budget chosen at kickoff. It can stop, request expertise or mark a result inconclusive. Save human feedback as a versioned research input; lock physical and budget constraints. It cannot change simulator code, invent observations or convert a diagnostic assumption into a recommended policy.

Completion target: one experiment and evaluation, a next step chosen by the agent, and visible human redirection. Execute the follow-up too if runtime permits. A negative or inconclusive result still demonstrates useful research.

PROJECT PLAN / 07

Architecture and runtime

One simulation engine and one neighbourhood scene serve residents and research

Component	Responsibility
React + TypeScript	3D street-level & aerial MapLibre neighbourhood view (buildings_3d.geojson), date/time shadow comparison (shadows.geojson), accessible 2D list, scenario controls, streetcar replay and research history.
Express / FastAPI + worker	Validate scenarios and experiments; queue CPU-heavy simulation jobs; expose status, results, grounded AI explanations and brief export. A job table and polling are enough.
Python model + prep	Prepare local spatial/GTFS/massing/shadow inputs; run streetcar and passenger events headlessly; compute crowding, wait and journey metrics and conservation diagnostics.
PostgreSQL / Tiger Data	Version sources, scenarios, runs and investigations. Tiger Data with PostGIS is verified live; local persistence fallback keeps the demo resilient.
Gemini + Papiers adapter	Gemini proposes structured hypotheses, scenario patches and grounded explanations. The adapter maps verified Papiers operations into the research workflow; set its contract after the access test. [9]

Shared contract before splitting work

Freeze one sample scenario, result and experiment record (Contract 2.0.0 on westbound 505 Dundas): site/stop/link IDs, coordinates, units, phase, demand profile, intervention IDs, budgets, seed configuration, metric definitions and status. A hypothesis record includes its source context, controlled changes, evaluation rule, run IDs, finding, human feedback and next step.

The 3D views, 2D list, replay, export and AI explanations all consume the same backend results and reviewed geometry. Every displayed number traces to a run or source. Save exact prepared inputs, model version and random draws; cache by the complete validated configuration. Retain aggregate metrics for all runs and detailed activity events for selected replays. Keys stay on the backend.

OPEN F2 | Benchmark before promising instant results
Engineering targets, not measured claims: a selected-phase central-demand four-package preview within 10 seconds; a full two-phase, three-case, 20-run matrix within 60 seconds on the demo host. That matrix is 480 runs before baselines, delay tests or research. Measure it early, show jobs asynchronously and label previews with run count. Never use a preview alone to certify readiness.

Reduce work without changing the evidence

Compute only the selected phase/case for exploratory previews; cache identical results; precompute flat-ground shadow layers and generate detailed replay events for one run. The research agent reuses the same worker and cache. Retain the full declared comparison for the final brief.

If database or hosting setup stalls, use a simple local persistence fallback and disclose which sponsor integrations are absent. If Papiers access fails, keep the resident neighbourhood and transit product usable while resolving access.

PROJECT PLAN / 08

Build plan and completion gates

Integrate early; name the owners at kickoff

Owner slot	Primary deliverable and coordination
A / Interface	3D street/aerial views, sunlight/shadow comparison, accessible 2D list, comparisons, streetcar replay and research panel. Help D with integration and own deployed-flow verification.
B / Data + demand	Reviewed 505 Dundas corridor bundle, 239/500/761 Dundas St E proposals, expanded building inventory, 3D massing, solar/shadow layers and package inputs. Pair with C on baseline/boundaries.
C / Simulation	Streetcar and passenger events, crowding/wait/journey metrics, invariants and experiment runner. Own hand-checkable tests and runtime measurement.
D / Research + API	Papiers access, agent loop, Express/FastAPI endpoints and Tiger Data persistence. A shares integration/export so this role does not absorb every backend task.

Gate	Evidence required before expanding scope
First 2-3 hours	B freezes the 505 westbound corridor, proposals, massing and shadows; C runs a tiny correct model; D proves Papiers/Gemini/Tiger Data access; A renders the shared sample.
Basic loop	A real 505 Dundas data bundle flows through the API to current/growth results, 3D proposal toggles and shadow views in the interface. Conservation and overloaded-queue tests pass.
Comparison + research	Four packages enforce resources; stop and journey metrics agree. One real Papiers-informed experiment completes, is evaluated, and generates a next step.
Robustness + demo	Complete declared phases/cases, street-level walk view, shadow comparison, delay test, uncertainty, human redirection, saved brief and replay. Benchmark on the actual host.
Final reserve	Freeze features at least four hours before the team's working deadline. Verify after reload, rehearse the resident story, submit repository link and confirm selected categories.

OPEN F7 | Named ownership and conflicting times
Assign A-D, one integration lead and one submission owner. Official Devpost pages conflict: the overview deadline is Oct 18 at 8:00 AM EDT; rules say 10:00 AM ET, with an initial submission Oct 17 at 9:00 PM ET. Supplied notes also differ on the start time. Confirm with organizers; plan against the earlier final time until resolved. [1]

Verification that earns completion

Check passenger conservation, streetcar capacity, ordered events, persistent denied riders and a deliberate overload. Verify reproducibility, phase/fleet/budget rejection, matched comparison cohorts, shadow layer alignment and unresolved-wait handling. Check that 3D/2D views, replay, export and AI numbers agree. Test one human correction changing the agent's next configuration; have a new user explain the selected package and its limitations.

If late: remove extra 3D facade polish and scrubbing first, then extra research cycles and optional writeback. Preserve the connected 3D/shadow/commute workflow, a genuine Papiers read, one completed evaluation and an agent-generated next step.

PROJECT PLAN / 09

Value, differentiation and fit

Resident-centred transit simulation connected to housing scale, shadows and AI research

Existing approach	Overlap and Urbanova's emphasis
VU.CITY / ArcGIS Urban [13,14]	Already provide 3D proposal visualisation, public urban models and shadow analysis. Urbanova connects street-level massing and seasonal shadows directly to a resident streetcar commute simulation.
CityCast [10]	Already combines development, household and transit changes with transport simulation. Urbanova emphasizes resident commute impacts, move-in phases, affordable service packages and an inspectable research trail.
UrbanSim [11]	Already connects development timing, land use and transportation scenarios. Our narrower product focuses on neighbourhood sunlight and the streetcar service available when specific housing opens.
Replan [12]	Already combines transit planning, costs, vehicles and simulated demand. Our emphasis is resident-centred housing readiness plus Papiers-informed investigations and visible human redirection.

3D proposal visualisation, shadow analysis and transport modelling already exist. We describe Urbanova's contribution through the connected resident experience: helping residents see how new housing could change their commute and neighbourhood, test what better transit could achieve, and bring understandable evidence into community discussions.

Tracks and themes

Target	Fit and what the demo must prove
School of Cities / primary	Clearest fit: help residents understand a concrete Toronto urban issue on 505 Dundas using real proposals, 3D massing, seasonal shadows and transit data with responsible interpretation. [1]
Papiers / primary	Actual autonomous investigation, evaluation and next step; real Papiers use; a human note or constraint visibly changes the research trajectory. Conditional on execution and access. [1,2]
AI for Social Impact	Help residents and community groups ask questions, understand commute/neighbourhood tradeoffs and challenge assumptions without claiming measured policy impact.
Climate Action	Investigate transit service supporting compact housing growth and explore seasonal sunlight/shade. This prototype does not establish avoided car trips, carbon savings or thermal comfort.
Gemini / secondary	Use real API calls for structured experiment planning, scenario parsing and grounded result explanations, validated by the backend. [1,9]
Tiger Data / Vultr	Enter only if actually used: inspectable saved runs/queries in Tiger Data; a working hosted API or worker for Vultr/Railway. No unmeasured performance claims. [1]

Economic inclusion is a secondary discussion of service access, not a demonstrated fintech outcome. Healthcare, FDM and Solana are outside the build. Avoid extending the product merely to collect categories.

PROJECT PLAN / 10

Limits, demo and source key

Keep the evidence, assumptions and unfinished questions visible

Current limitations and responsible interpretation

F1: corridor is fixed to TTC 505 Dundas westbound (Broadview to TMU Station, 9 stops) with 239, 500 and 761 Dundas St E; observed stop-level demand, the 761 replacement unit count, and walking field review remain open. F2: runtime targets require host benchmarking. F4: demo thresholds and ranking priority are user-selected. F5: streetcar intervention effects, costs and fleet assumptions require explicit labels. F6: live Papiers access is gated on verification. F7: named owners and authoritative times require confirmation. F3 is addressed by journey-time and in-vehicle crowding metrics.

Urbanova is an exploratory scenario tool. It distinguishes proposed designs, simplified 3D masses and hypothetical changes; compares added and removed shadows with existing surroundings at the same date and time (without claiming thermal-comfort modelling); and presents transit outcomes as conditional scenarios with benefits and pressures shown together.

Four-minute resident demonstration

0:00-0:40: Find a familiar place on 505 Dundas & toggle proposed 3D towers (239, 500, 761 Dundas St E) from street level and above. 0:40-1:30: Explore seasonal sunlight/shadows (today, added, removed), then inspect the westbound streetcar commute and compare transit packages. 1:30-2:50: Ask why a downstream stop worsens; run the Papiers/Gemini experiment and redirect its next step with a human constraint. 2:50-3:30: Inspect the next configuration and phase/budget limits. 3:30-4:00: Save the comparison brief with sources and uncertainties for community discussion.

Sources and input material

Public pages checked October 16-17, 2026. Links establish source availability; downloads, calibration and integration access are separate build gates.

[1] Hack the Valley 11 overview and rules | Rules
[2] Papiers Autoresearch Challenge (user-supplied full brief); product
[3] City of Toronto: Development Pipeline & Application Information Centre (239, 500, 761 Dundas St E)
[4] Toronto Open Data: Merged GTFS TTC Routes and Schedules (Route 505 Dundas)
[5] TTC: 2025 Operating Statistics & Transit Planning (32,100 weekday 505 boardings)
[6] OpenStreetMap contributors (ODbL) pedestrian network & building footprints
[7] Toronto Open Data: 3D Massing & NOAA Solar Position / Flat-Ground Shadow Equations
[8] TTC: RapidTO Surface Transit Network Plan & streetcar priority assumptions
[9] Google: Gemini structured outputs (@google/genai)
[10] CityCast: public product description
[11] UrbanSim: Scenario Modeling
[12] Replan: Transit and SIM
[13] VU.CITY: 3D urban planning and shadow analysis (https://www.vu.city/planners)
[14] ArcGIS Urban: 3D scenario modeling (https://learn.arcgis.com/en/projects/explore-an-urban-model/)

Also used: docs/doc/project_desc.md, docs/simulation-data.md, docs/massing.md, docs/contract.md and the Urbanova concept/build brief.